

# Kata.ai: Creating natural language processing solutions to serve 200 million+ Indonesians

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[About Kata.ai](#)



## About Kata.ai

Kata.ai built an e-grade conversational platform to power assistants for companies in Indonesia, where multiple languages are spoken including the classical language of Java

Kata.ai is an Indonesian conversational AI company that uses natural language processing (NLP) to power virtual assistants for major corporations in Indonesia across different industries, including fast-moving consumer goods, financial services, telecommunications, and retail. Launched in 2016, it's on a mission to "elevate people's potential through the power of language and technology."

Industries: Technology

## Google Cloud results

- Enabled natural language processing researchers to crack one of the world's toughest languages, Bahasa Indonesia
- Yielded 40% cost savings month-on-month through productivity gains
- Solved persistent stability problems with Kubernetes cluster by switching to Google Kubernetes Engine
- Opened new business

Goog  
Kuber  
Engin  
produ  
50%

Location: Indonesia

Cloud Speech to Text

Compute Engine

(<https://cloud.google.com/compute/>)

Dataflow (<https://cloud.google.com/dataflow/>)

Cloud Speech to Text

(<https://cloud.google.com/speech/>)

Cloud SQL (<https://cloud.google.com/sql/>) help.

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

opportunities  
with data  
center  
deployment  
plans

Indonesia has 701 languages

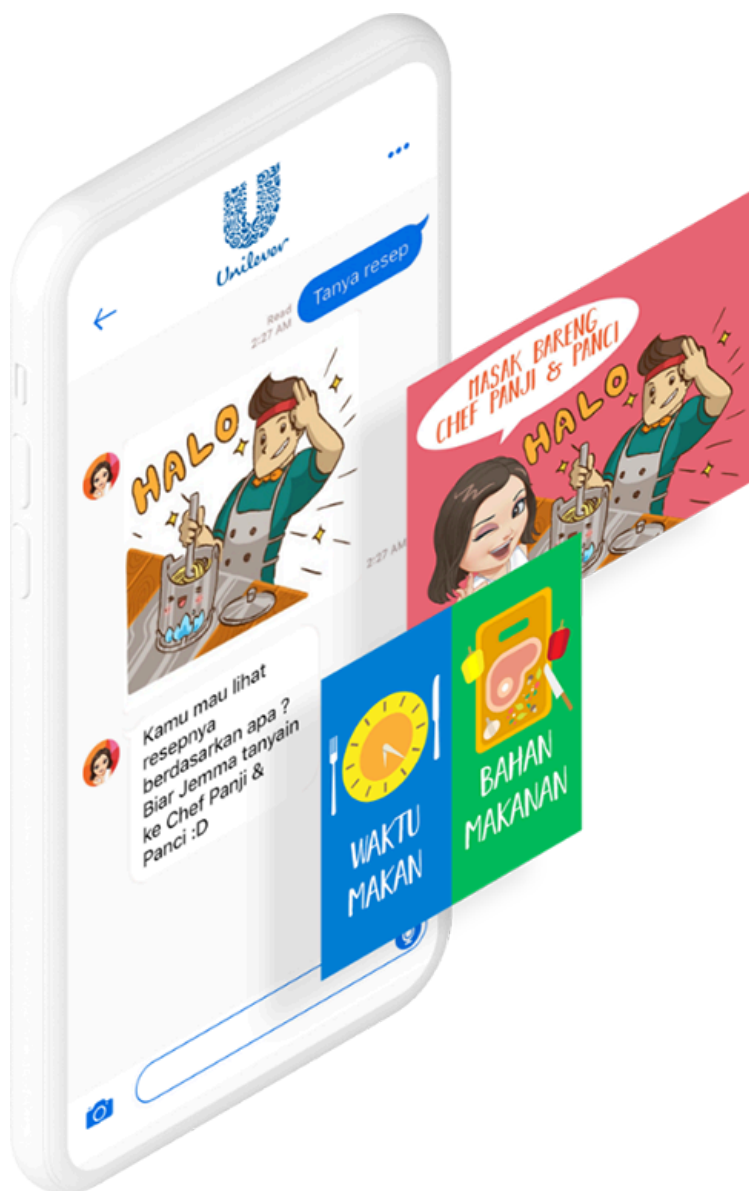
(<https://www.sciencedirect.com/science/article/pii/S2352711018302413>)

spoken among 200 million people. In fact, the official language, Bahasa Indonesia, is spoken at home by just one-fifth of the population. And when used in public, it comes in an endless variety of dialects, with lots of slang thrown in. That's what makes Kata.ai (<https://kata.ai/>) so unique.

A conversational AI startup, Kata.ai develops enterprise-grade natural language processing (NLP)-powered virtual assistants specifically for Indonesian corporations, from financial groups to telecom operators. The solutions its platform can provide range from marketing that targets Indonesian teens using local lingo, to allowing telecom customers to top up smartphone data via chatbots in various messaging platforms.

"Developing a system that can master Indonesian languages is our biggest challenge," says Kata.ai CTO Pria Purnama. To tackle it, Kata.ai assembled a team of Indonesian NLP researchers and linguists spanning Jakarta, Alabama, and Edinburgh, Scotland, to develop an AI engine capable of navigating Indonesia's linguistic diversity.

To succeed, Kata.ai needed a powerful and stable cloud infrastructure enabling researchers to create and collaborate, share, and test data, across borders and timezones. The cloud provider it initially tapped led to persistent stability issues on its Kubernetes (/kubernetes) cluster. After giving Google Cloud (<https://cloud.google.com/>) a test drive, Kata.ai saw its platform's connectivity problems disappear virtually overnight. The results opened a treasure chest of exciting new possibilities.



## Bedrock for AI language construction

Because Indonesia has so many different languages and dialects, naturally there is limited language documentation for NLP researchers to model. On top of that, Pria says that because Indonesian NLP is such a niche area, specialists can be very hard to find.

These realities compelled Kata.ai to scour the world for top-notch experts in obscure corners of R&D and academia. The glue for this far-flung team had to be the most powerful and reliable cloud computing service available. With Kata.ai's original provider, however, "the network would intermittently go down, and we couldn't understand why," says Pria. The team ended up wasting time on tasks such as fixing infrastructure issues, as well as manually configuring cloud computing services.

The issue went beyond lost manpower hours: "When network breakdowns happen," says Pria, "it just breaks the flow of creativity."

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—Pria Purnama, CTO,  
*Kata.ai*

Kata.ai reached out to Google Cloud for a test run, out of a hunch that its advanced Kubernetes solutions might provide the solid foundations it needed for building its Indonesian-language "castle in the sky." "We rely on Kubernetes to orchestrate and deploy," says Pria. "As we're a distributed team, with engineers all over the globe, Kubernetes is pivotal to our workflow and collaboration. A stable solution here is mission critical."

Kata.ai ran its proof of concept on a subset of Kata.ai's entire platform, as it wanted to see first whether its Kubernetes connectivity problems persisted with Google Cloud: "When no issues came up and everything ran smoothly, we decided to move forward with a phased migration."

"The experience proved our theory that the network on Google Cloud is simply more stable," says Pria. "Google Kubernetes Engine (/kubernetes-engine) is a much more mature product compared to our previous provider. We literally rest better at night knowing the pods are not going to disconnect with each other. And as a direct result, we're free to do more productive and creative tasks."

The numbers? An estimated 50% boost to productivity and 40% cost savings from improved efficiencies: "The key thing about Google Cloud that won us over was network stability," says Pria. "We spent so much time with our previous provider trying to make this work, and ultimately we decided the way to go was with Google Cloud."

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## **A proactive partner**

It's not only network reliability and computing tools that triggered Kata.ai's decision. In Google Cloud, Kata.ai also discovered a partner willing to go the extra mile to help it solve difficult problems. It was a change from the previous provider, which Pria and his team viewed as having a less open and collaborative approach.

Google Cloud offered personalized assistance, brainstorming sessions, and CTO roundtable



discussions to help Kata.ai unlock the full potential of the solutions available.

“With Google Cloud, when we had a question, we would just pick up the phone and call,” says Pria.

“The team buckled down together with us to find a way to make things work. This level of support is one of our deciding factors for moving forward with Google Cloud.”

For a complex AI platform, switching cloud providers isn't a decision to be taken lightly: “Migration is not easy, because we have to keep our solution running without interruption, which means we can't risk prolonged downtime,” says Pria. “We felt confident about the move because the Google Cloud team has proven, time and time again, that it would be there for us when we need it. That tipped the balance.”

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## **Future growth with Indonesian data center and NLP solutions**

As Google Cloud continues to expand its technical infrastructure network globally, Pria is looking forward to its cloud region in Indonesia in 2020 (<https://techcrunch.com/2019/04/09/google-is-bringing-two-new-data-centers-online-in-2020/>)

. This will mean that new Indonesian compliance restrictions on moving data beyond national borders won't be a problem for Kata.ai. Instead, it will be a major step toward allowing Kata.ai to provide more solutions to clients, especially those who carry out on-premises deployment, such as banking, financial services, and insurance (BFSI) companies.

"We look forward to a Google Cloud data presence in Indonesia," says Pria. "This is going to enable us to explore a wealth of exciting possibilities for our platform and our customers."